



Assessment of the Australian Bureau of Meteorology wet bulb globe temperature model using weather station data

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Abstract

Exertional heat illnesses affect thousands of athletes each year and are a leading cause of death in sports. The wet bulb globe temperature (WBGT) is widely used as a heat stress metric in athletics for adjusting activities. The WBGT can be measured on-site with portable sensors, but instrument cost may provide a barrier for usage. Modeling WBGT from weather station data, then, presents an affordable option. Our study compares two WBGT models of varying levels of sophistication: the Australian Bureau of Meteorology (ABM) model which uses only temperature and humidity as inputs and a physically based model by Liljegren that incorporates temperature, humidity, wind speed, and solar radiation in determining WBGT outputs. The setting for the study is 19 University of Georgia Weather Network stations selected from across the state of Georgia, USA, over a 6-year period (2008–2014) during late summer and early fall months. Results show that the ABM model's performance relative to the Liljegren model varies based on time of day and weather conditions. WBGTs from the ABM model are most similar to those from the Liljegren model during midday when the assumption of moderately high sun most frequently occurs. We observed increasingly large positive biases with the ABM model both earlier and later in the day during periods with lower solar radiation. Even during midday, large (≥ 3 °C) underestimates may occur during low wind conditions and overestimates during periods with high cloud cover. Such differences can lead to inaccurate activity modification and pose dangers for athletes either by underestimating heat-related hazards or by imposing an opportunity cost if practice activities are limited by overestimating the heat hazard.

Keywords Heat stress · Athletics · Wet bulb globe temperature · Model

Introduction

Exertional heat illnesses (EHIs) affect thousands of athletes each year and are a leading cause of death among athletes (Kerr et al. 2013; Casa et al. 2012). EHIs are complex and associated with both intrinsic (e.g., health, hydration, acclimatization) and extrinsic (e.g., weather, equipment, work-to-rest ratios) factors (Casa et al. 2015). Weather is an important extrinsic factor that can be used in planning practice activities so that they are appropriate for the conditions. Within the athletic community as well as the military and occupational settings, the wet bulb globe temperature (WBGT) is

commonly used to guide activity modification (e.g., NIOSH 2016; Armstrong et al. 2007; Casa et al. 2015; Department of Army and Air Force 2003). The WBGT is an environmental measure that integrates the influences of air temperature, humidity, wind speed, and radiant heating via a weighted average of the globe (GT), wet bulb (WB), and dry bulb (DB) temperatures in the following equation (Yaglou and Minard 1957):

$$\text{WBGT} = 0.7 \times \text{WB} + 0.2 \times \text{GT} + 0.1 \times \text{DB}$$

Other widely available metrics for assessing heat stress such as the heat index or the UTCI are not suitable for application to athletic activities. These indices assume, among other factors, that individuals are only moderately active (e.g., walking) and may therefore underestimate the heat stress of an athlete with a high metabolic rate (Bröde et al. 2012; Rothfucz 1990).

Ideally, WBGT is measured on-site to provide the best representation of exposure conditions. Indeed, some

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organizations like the Georgia High School Association (GHSA) explicitly require on-site measurements in their heat policy to guide practice and training activities (GHSA 2018). Unfortunately, the cost of handheld WBGT sensors may pose an impediment for use, especially among volunteers who work in youth sport activities. As such, being able to get WBGT values from weather stations is an appealing and affordable option.

WBGT can be estimated from weather station data with models of varying levels of sophistication (Lemke and Kjellstrom 2012). A simple, commonly used model is available from the Australian Bureau of Meteorology (hereafter referred to as the ABM model; ABM 2018). It relies solely on air temperature and humidity, and the ABM provides a chart where a user can determine the approximate WBGT from temperature and relative humidity. Because it can be computed from commonly available weather data or model fields, it has been used in research studies of athletes and laborers where WBGT data were not collected in-situ (e.g., Smith et al. 2017; Kakamu et al. 2017; Cooper et al. 2016; Altinsoy and Yildirim 2015; Brito et al. 2014; Willet and Sherwood 2012; Grimmer et al. 2006). This model, however, has been criticized as an incomplete representation of WBGT because it does not account for the influences of the globe temperature (Budd 2008). The ABM indicates that this model can underestimate when the assumptions of moderately high sun and low wind speeds are not met (ABM 2018). Yet, there is no quantification of the magnitude of these biases and how they might impact activity modification of athletes or others relying on these WBGT estimates. More sophisticated models such as by Liljegren et al. (2008) account for the full suite of meteorological data in computing WBGT and have been documented to be highly accurate (Liljegren et al. 2008). The objective of this study is to compare WBGT estimates from the ABM model versus a highly sophisticated WBGT model over a range of weather conditions at multiple weather stations across the state of Georgia, USA.

Data and methods

Two different WBGT models were used in this study. The ABM model uses an approximation of the WBGT that is based on an equation published in 1984 in an American College of Sports Medicine (ACSM) Position Stand (ACSM 1984). This model considers only temperature and humidity in the following equation:

$$\text{WBGT} = 0.567 \times T_a + 0.393 \times e + 3.94 \quad (1)$$

where T_a is the dry bulb temperature ($^{\circ}\text{C}$) and e is the water vapor pressure (hPa). It does not use data on solar radiation or wind speed but rather assumes a “moderately high radiation

level in light wind conditions” (ABM 2018). As mentioned earlier, the ABM notes that the WBGT approximation may under- or overestimate outside of the assumed conditions, particularly if the weather is cloudy or during times such as early morning or nighttime when the sun is low or below the horizon (ABM 2018).

A second model by Liljegren et al. (2008; hereafter referred to as the “Liljegren model”) is a sophisticated, mathematical model. It requires meteorological data on air temperature (dry bulb temperature), relative humidity, solar radiation, and wind speed to calculate the wet bulb (WB) and globe temperatures (GT). The WB and GT are combined with measured dry bulb (DB) temperature to compute the WBGT. The natural wet bulb temperature (T_w) is determined based on the energy balance of a wetted wick. Air temperature (term 1, Eq. 2) is combined with terms that consider energy losses via evaporation (term 2, Eq. 2) and energy gains via radiation and convection (term 3, Eq. 2) in the following equation:

$$T_w = \underbrace{T_a}_{\text{Term 1}} - \underbrace{\frac{\Delta H}{C_p} \frac{M_{\text{H}_2\text{O}}}{M_{\text{Air}}} \left(\frac{Pr}{Sc}\right)^a \left(\frac{e_w - e_a}{P - e_w}\right)}_{\text{Term 2}} + \underbrace{\left(\frac{\Delta F_{\text{net}}}{A h}\right)}_{\text{Term 3}} \quad (2)$$

where T_a is ambient air temperature, ΔH is the heat of vaporization, C_p is the specific heat at constant pressure, $M_{\text{H}_2\text{O}}$ is the molecular weight of water vapor, M_{Air} the molecular weight of air, Pr is the Prandtl number, Sc is the Schmidt number, a is a constant = 0.56, e_w is the saturation vapor pressure of the wick, e_a is the saturation vapor pressure of the air, P is the barometric pressure, ΔF_{net} is the net radiant heat flux from the environment to the wick, A is the surface area of the wick, and h is the convective heat transfer coefficient which is computed in part from wind speed.

The globe temperature (T_g) is computed from the energy budget of the black globe. In Eq. 3, term 1 represents exchanges of thermal radiation from the air, term 2 accounts for convective energy transfers, and term 3 represents energy gains by the globe from solar radiation:

$$T_g^4 = \underbrace{\frac{1}{2}(1 + \varepsilon_a)T_a^4}_{\text{Term 1}} - \underbrace{\frac{h}{\varepsilon_g \sigma}(T_g - T_a)}_{\text{Term 2}} + \underbrace{\frac{S}{2\varepsilon_g \sigma}(1 - \alpha_g) \left[1 + \left(\frac{1}{2\cos(\theta)} - 1 \right) f_{\text{dir}} + \alpha_{\text{sfc}} \right]}_{\text{Term 3}} \quad (3)$$

where ε_a is emissivity of the air, T_a is air temperature, h is the convective heat transfer coefficient, ε_g is the emissivity of the globe, S is the total horizontal solar irradiance, α_g is albedo of the ground, θ is the solar zenith angle, f_{dir} is the fraction of S

that is direct beam radiation, and α_{sfc} is the albedo of the surface. We set $\varepsilon_g = 0.95$, $\alpha_g = 0.05$, and $\alpha_{\text{sfc}} = 0.45$ as used in Liljegren et al. (2008). Iterative numerical solutions were used to solve the wet bulb and globe temperature equations.

The Liljegren model is arguably the best WBGT model available. Lemke and Kjellstrom (2012) note it meets best standards for a WBGT model as it uses a full suite of weather variables (e.g., air temperature, humidity, solar radiation, and wind speed) for input data, performs heat exchange computations based on thermodynamic principals, and has been well tested. Indeed, the model has been found to be accurate to within 1 °C at diverse study sites with different climates (Liljegren et al. 2008) and outperformed other WBGT models during field testing (Lemke and Kjellstrom 2012; Patel et al. 2013). Further, the 1 °C difference between the Liljegren model and observed WBGTs is similar to the reported variability among commercial WBGT sensors (Cooper et al. 2017). Given its well-documented ability to simulate measured WBGTs, we considered the Liljegren WBGTs as “ground truth” for comparison purposes.

Input meteorological data for the models were acquired from the University of Georgia Weather Network (UGAWN). This station network collects a suite of meteorological data at 15 min intervals (UGAWN 2018). For this project, we used air temperature (dry bulb temperature), solar radiation, wind speed (2 m), and humidity data (relative humidity and dew point temperature). We selected data for the months of August and September from 19 stations spread across the nine US climate divisions of Georgia over a 6-year period (2008–2014; Fig. 1). The choice of stations and time range coincides with the use of these data in a separate study of Georgia High School football players and exertional heat illnesses that was conducted over the same time period (Grundstein et al. 2017). The late summer and early fall months are a particularly relevant time for the study as it is a hot and humid time in Georgia and coincides with the beginning of fall interscholastic sports such as football (NCEI 2017). The 19 weather stations were selected based on their close proximity to the specific high schools participating in the study (Fig. 1). In total, over 650,000 observations were recorded and used as model input. These data represent a wide range of weather conditions which will allow an assessment of the sensitivity of WBGT output from each model to changing weather (Table 1). In particular, solar radiation varied from zero to almost 1100 W m⁻², temperature ranged from a cool 3.5 °C to the hot 37.8 °C, wind speeds differed from zero to over 8 m s⁻¹, and dew point temperatures varied from dry at –2 °C to very humid at 27.8 °C.

The same meteorological data were used as inputs for simulations with both models. For analysis, the WBGT outputs were aggregated by hour. This approach was used in light of statements by the ABM that the model is optimal during periods of high sun (midday) and less effective during periods of

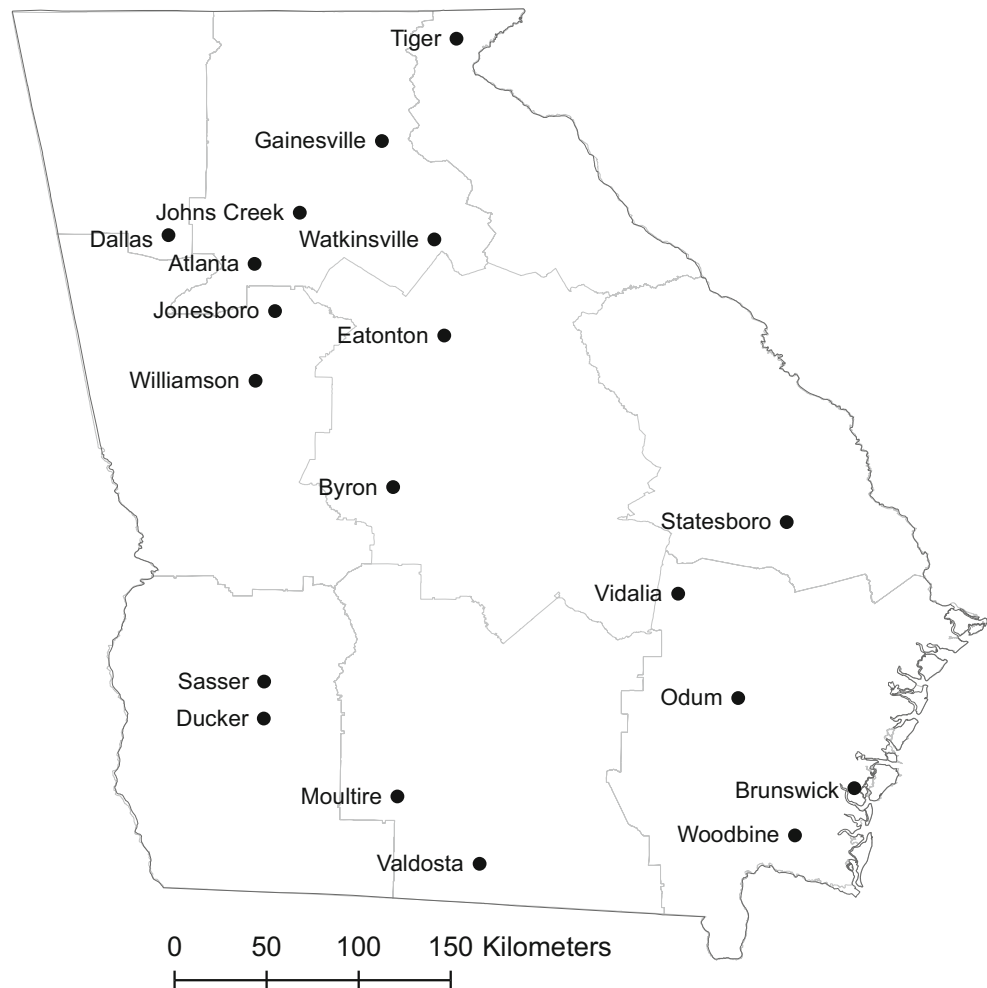
low sun (morning and evening). We also assessed how varying weather conditions in given hours may lead to differences in estimated WBGT between the models. In the analysis, we considered –1 to 1 °C differences to be *good agreement* as this is within the resolution of handheld WBGT sensors (Cooper et al. 2017). We also considered differences ≥ 3 °C (“positive bias”) and ≤ -3 °C (“negative bias”) as *poor agreement*, representing a little more than 1 standard deviation from the mean difference during the 11 a.m. to 1 p.m. time frame. Lastly, we examined how the two models differed during highly oppressive weather as defined by the spatial synoptic classification system (Sheridan 2002). An air mass based approach allows us to consider the combination of multiple variables (e.g., air and dew point temperatures, wind speed, cloud cover) that might affect heat stress. We focused on the moist tropical plus (MT+) air masses which are unusually hot and humid, and have been associated with increased heat-related mortality (Sheridan and Kalkstein 2004). There were 2 days in the dataset (August 18 and 22, 2010) where an MT+ air mass was present at most (75%) weather stations across the state (Sheridan 2018).

Finally, we consider the WBGTs in the context of activity modification guidelines. As mentioned earlier, specific guidelines have been developed for athletic, occupational, and military settings (e.g., Armstrong et al. 2007; NIOSH 2016; Department of Army and Air Force 2003). In this study, we used WBGT-based activity modification from the American College of Sports Medicine (ACSM; Armstrong et al. 2007) which is widely adopted in the athletic community. The ACSM provides different guidelines based on activities (e.g., continuous activity such as cross country running or non-continuous activity as represented in American Football) or the health and fitness of the athlete. The categories listed in Table 2, which are commonly emulated in American Football heat safety policies, provide an example of these guidelines for fit, acclimatized athletes participating in non-continuous activity. As the WBGT increases, the guidelines indicate greater caution and limits to activity. For instance, a WBGT of 31 °C would require limits to intense exercise and total practice time but a WBGT of 32.3 °C, one category higher, would require that the practice be canceled. In our analysis, we will consider the practical implications of differences in WBGTs between the models using the categories listed in Table 2.

Results

The difference in WBGT between models (ABM—Liljegren) was examined by hour using box plots (Fig. 2). For each hour, 27,128 values were compared allowing us to investigate a wide range of weather conditions during the August–September study period. In Fig. 2, positive (negative)

Fig. 1 Locations of University of Georgia Weather Network (UGAWN) stations used in the study. Light gray lines show U.S. Climate Division boundaries



WBGD differences indicate the ABM WBGDs are greater (less) than those produced by the Liljegren model. On average, the ABM model has a positive bias. This bias is smallest between 10 a.m. and 2 p.m. with median differences ranging from 1.2 to 1.6 °C. The magnitude of positive bias increases in the morning and afternoon hours with median differences between 2 and 3 °C and is highest (4–5 °C) at low sun angles near dawn and dusk and during nighttime hours. Importantly, the box plots show a wide range about the median differences, especially from 7 a.m. to 5 p.m., with the 25th and 75th percentiles spread across a little less than 2 °C and the extremes of

the whiskers (the 75th percentile + $1.5 \times$ interquartile range and 25th percentile – $1.5 \times$ interquartile range) spanning over 6 °C.

We investigated the meteorological conditions that led to wide differences between the models during the period from 11 a.m. to 1 p.m. ($n = 81,384$ observations) when the median differences were smallest and the ABM model performs best (Table 3). Almost 30% ($n = 24,328$) of model differences were between –1 and 1 °C; about 10% were ≥ 3 °C ($n = 8188$) and less than 1% were ≤ -3 °C ($n = 514$). We used z scores, representing the number of standard deviations from the

Table 1 Weather statistics during 6-year study period

	Wind speed (ms^{-1})	Solar (Wm^{-2})	Temperature (°C)	Dew point (°C)
avg	1.28	206.27	24.27	19.78
stdev	0.90	280.61	4.63	3.45
max	8.29	1098.57	37.81	27.80
min	0.00	0.00	3.47	–2.09
count	651,071	651,071	651,071	651,071

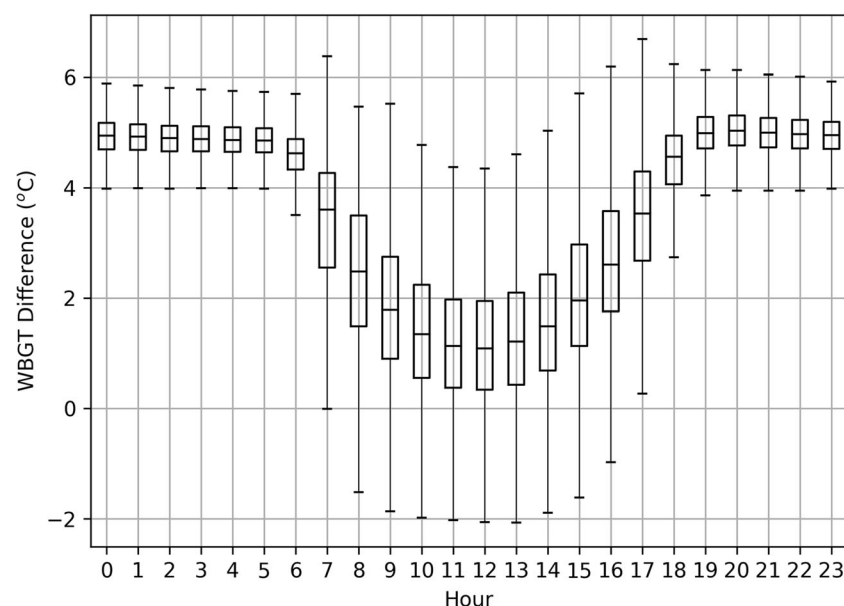
Table 2 American College of Sports Medicine (ACSM) WBGT activity modification guidelines for acclimatized, fit, and low-risk individuals participating in non-continuous activities (Armstrong et al. 2007)

	WBGT (°C)	Activity modification
1	≤ 10.0	Normal activity.
2	10.1–18.3	Normal activity.
3	18.4–22.2	Normal activity.
4	22.3–25.6	Normal activity. Monitor fluid intake.
5	25.7–27.8	Normal activity. Monitor fluid intake.
6	27.9–30.0	Plan intense or prolonged exercise with discretion; watch at-risk individuals carefully.
7	30.1–32.2	Limit intense exercise and total daily exposure to heat and humidity; watch for early signs and symptoms.
8	> 32.3	Cancel exercise uncompensable heat stress exists for all athletes.

mean, to assess whether a particular meteorological value was typical (about average) or extreme. Periods with good agreement were characterized by average or near-average wind speeds, dew point temperatures, and air temperatures, with z scores between -0.5 and 0.5 . Solar radiation was slightly above average with a z score of 0.56 , which is consistent with the ABM's statement that it is designed for "moderately high" radiation levels. The positive bias conditions where the ABM overestimates WBGTs relative to the Liljegren model were associated with very low solar radiation (z score = -1.82) and air temperatures which is likely correlated to the low solar radiation. Negative bias occurs during periods with very low wind speeds which are over two standard deviations from the mean (z score = 2.08). Finally, we assessed how the models compared during two days with unusually oppressive weather as indicated by the presence of an MT+ air mass across the

state. These days were very hot and humid with average air and dew point temperatures of approximately 31.6 °C and 23.2 °C, respectively, and just under 1 standard deviation from the mean. Solar radiation was slightly above average and wind speeds slightly below average. Both Liljegren and ABM models indicated high WBGTs, averaging about 31.4 °C and 33.0 °C respectively, with the ABM model showing a positive bias of about 1.6 °C. The ABM model on these days falls slightly outside of the "good agreement" category, mostly likely because the solar radiation is not sufficiently high enough and the wind speeds not low enough to meet the ABM model assumptions.

The wide differences that may occur between models depending on the weather conditions may have substantial impacts on activity modification. Using the ACSM activity modification guidelines (Armstrong et al. 2007), the categorical

**Fig. 2** WBGT difference between the ABM and Liljegren models. A positive (negative) difference indicates that the ABM model provided a higher (lower) WBGT estimate. The bottom and top of the box represent

the 25th and 75th percentiles, the line inside the box is the median, and the whiskers represent the 75th percentile + $1.5 \times$ interquartile range and 25th percentile $- 1.5 \times$ interquartile range, respectively

Table 3 Meteorological conditions representing different levels of agreement between models from 11 a.m. to 1 p.m.

		Wind (ms ⁻¹)	Solar (Wm ⁻²)	Temperature (°C)	Dew point (°C)
Good agreement (− 1 to 1 °C)	Avg	1.75	781.54	28.38	18.12
	Stdev	0.58	143.33	3.21	3.88
	n	24,328.00	24,328.00	24,328.00	24,328.00
	zscore	− 0.25	0.56	0.05	− 0.44
Positive bias (≥ 3 °C)	Avg	2.02	192.46	25.16	21.65
	Stdev	0.93	112.65	3.95	2.34
	n	8188.00	8187.00	8188.00	8188.00
	zscore	0.10	− 1.82	− 0.83	0.52
Negative bias (≤ − 3 °C)	Avg	0.31	768.13	27.82	17.83
	Stdev	0.16	133.37	3.13	3.80
	n	514.00	514.00	514.00	514.00
	zscore	− 2.08	0.51	− 0.10	− 0.52
MT+ aimass	Avg	1.85	720.65	31.59	23.16
	Stdev	0.64	181.91	1.58	1.57
	n	216.00	216.00	216.00	216.00
	zscore	− 0.12	0.32	0.94	0.92

differences were investigated considering 0, 1, and ≥ 2 category differences in a stacked bar chart (Fig. 3). The smallest categorical differences occur during midday between 11 a.m. and 1 p.m. At noon, for instance, 49% of the estimated WBGTs had the same categorical designation, 46% has a one categorical difference, and about 5% had differences of two or more categories. The categorical discrepancies increase both earlier and later in the day. A morning practice at 8 a.m., for instance, would have categorical discrepancies over 75% of the time (i.e., one category—59% and two categories—16%, respectively) while at 3 p.m., about 71% would have a

discrepancy (i.e., one category—55% and two categories—16%, respectively).

The practical implications of these discrepancies are most relevant for categories 6–8 in Table 2, when activity restrictions are indicated. We examined “opportunity costs” when the ABM WBGT indicated categories 6–8, but the Liljegen WBGT indicated a category under 6, when no restrictions would apply (Fig. 4). Here, we found that 25% of the time, the use of the ABM would suggest restricted activity when none may have been needed. Most of these cases involved a category difference of two or more (79%), and the most

Fig. 3 Difference in American College of Sports Medicine (ACSM) activity modification category between ABM and Liljegen models

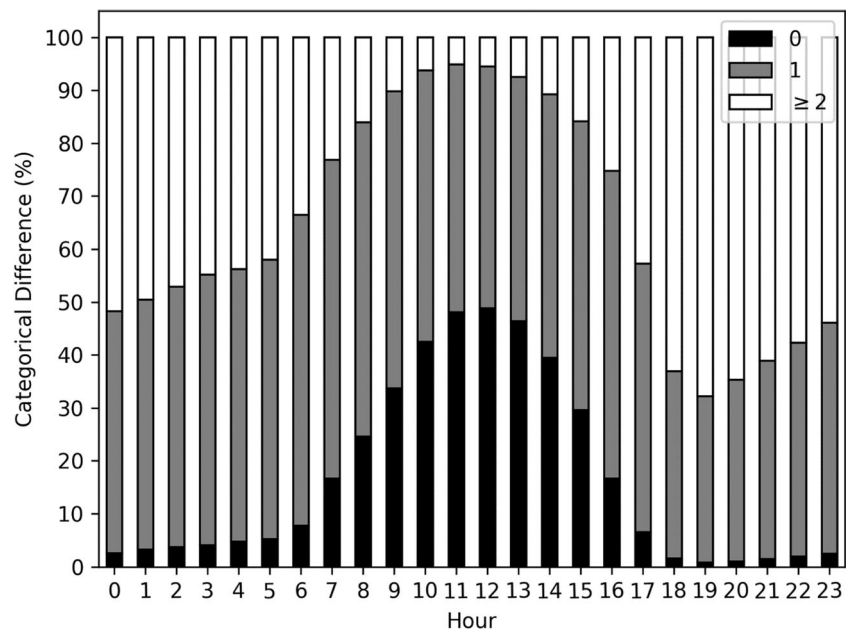
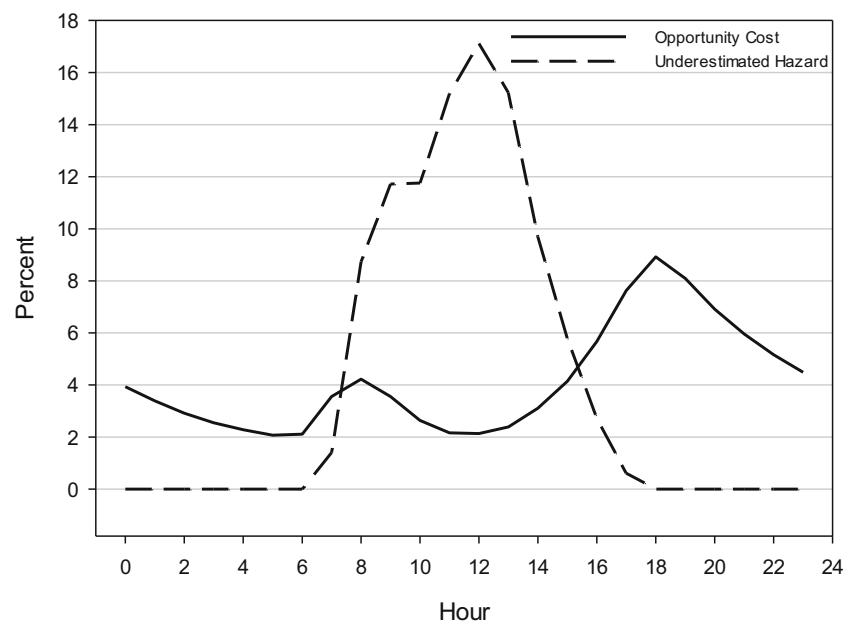


Fig. 4 Frequency in percentage of ACSM category differences between ABM and Liljegren WBGT models when opportunity costs (i.e., ABM would limit and Liljegren allow activity) or underestimated hazards (i.e., ABM would allow and Liljegren limit activity) may occur



frequent occurrences were in the late afternoon and evening (4–9 p.m.; Fig. 4). For example, in one set of 7 a.m. weather observations, the Liljegren WBGT (26.4 °C) indicated category 5 and the ABM WBGT (30.7 °C) indicated a category 7. Based on the Liljegren WBGT, there would be no change to practice activity but using the ABM WBGT, practice length, and intensity would need to be reduced. In addition, we considered “underestimated hazards” when the Liljegren WBGT indicated categories 6–8 activity modifications but the ABM WBGT was under 6 and allowed normal activity. For example, one particular set of weather observations from 1:30 p.m. is associated with a Liljegren model WBGT = 29.11 °C (category 6) which would require discretion in planning activities while the ABM model WBGT = 25.22 °C (category 4) would require no activity modification. This condition was much less common and occurred in only 0.35% of our observations. The vast majority of discrepancies were one category (82%) and tended to occur most often midday (Fig. 4).

Discussion and conclusions

We compared WBGT model output from a highly sophisticated mathematical model by Liljegren which uses multiple weather variables as inputs with the simpler ABM model which uses only dry bulb temperature and humidity as inputs. The ABM model is often used because the inputs are commonly measured at weather observing stations. This model tends to provide conservative values by overestimating WBGT, especially in the morning and evening. During average conditions during midday, it can provide WBGTs that are similar to those of the Liljegren model. However, it is important to

consider that weather is characterized by short-term variations in solar radiation, temperature, and wind speed and therefore is not always represented by average conditions. Indeed, we found that even during midday, when on average the ABM performs best, wide differences in WBGT can occur under certain weather settings. The ABM model will overestimate (≥ 3 °C) relative to the Liljegren model on days with high cloud cover that limits solar radiation and will underestimate (≤ -3 °C) on days with low wind speeds.

In not considering the variability of solar radiation and wind speed, then, the ABM model may inaccurately estimate WBGT and in turn lead to an improper recommendation for adjusting activity. We found at best, during midday, ABM model had a discrepancy of one or more categories in 50% of the time periods assessed; the discrepancies increase both earlier and later in the day with $\geq 70\%$ categorical discrepancies in the morning and afternoon/evening. This can put athletes at risk by not properly restricting activity as appropriate or may be an opportunity cost in lost practice time if it over-restricts activity when it is not truly necessary based on prevailing weather conditions. As the ABM model tends to provide conservative estimates, we found that the latter situation (e.g., opportunity cost) occurred more often at approximately 25% of the time, especially in the late afternoon and evening.

While our study focused on warm season weather in a hot-humid climate, we believe the findings are broadly applicable to other climates regions such as hot-arid ones. The ABM model is sensitive to variations in air temperature and humidity which would capture, for instance, the high temperature and low moisture conditions in a hot-arid location. The lack of sensitivity comes principally with solar radiation and how it may vary with cloud cover or low sun angles earlier or later in

the day. Regional factors may influence the relative impact of solar radiation on the ABM estimated WBGTs. For example, the role of cloud cover on solar radiation may become more pronounced in the arid Southwest USA during the monsoon season (late summer through early fall) when there may be increased daytime cloud cover (Adams and Comrie 1997).

Considering these findings, the ABM model is not recommended for estimation of WBGT for use in activity modification as it is not sensitive to changes in weather conditions. In the absence of on-site WBGT measurements, many states such as Georgia, North Carolina, Oklahoma, Delaware, and New Jersey among others maintain weather networks that provide sufficient data (e.g., air temperature, humidity, wind speed, solar radiation) to drive the more sophisticated WBGT models. A limitation of the Liljegren and other sophisticated WBGT models at present is that they are not easy for a coach or athletic trainer to use. In the short term, then, we hope our results will better inform clinicians about the issues of using the ABM WBGT estimates. In the near future, however, we hope to couple models such as the Liljegren model with data from weather networks to provide user-friendly, real-time WBGT estimates via outlets such as mobile apps.

For researchers who use WBGT retrospectively to study exertional heat injuries, there are a variety of datasets available to run models such as the one by Liljegren. The National Solar Radiation Database (NSRDB) that produced the National Renewable Energy Laboratory (NREL 2012) provides a serially complete dataset of meteorological variables, including solar radiation at locations throughout the USA and in selected regions internationally. This dataset has been previously used to run the Liljegren model to produce long-term times series of WBGT across the contiguous USA (e.g., Grundstein et al. 2014, 2015). Reanalysis datasets such as the second modern-era retrospective analysis for research and applications (MERRA-2) may also provide viable sources of input data (Gelaro et al. 2017). Hosokawa et al. (2018) for instance used MERRA-2 as input data to model WBGT for locations in Qatar and Japan.

In conclusion, we observed that the ABM WBGT model may not accurately represent heat hazards (specifically exertional heat illnesses) for those participating in outdoor activities. When on-site WBGT measurements are not available, we highly recommend the use of more sophisticated WBGT models with the increasing availability of suitable meteorological input data.

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